

Making AI Work: Tracing Human Labour in the Supply Chains of Dataset Production

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ABSTRACT

Data annotation is indispensable to AI development; annotation tasks ranging from image labelling to toxicity detection capture contextual human judgement needed to train and sustain AI systems. Yet it is often constituted as a repetitive, standardised activity performed by seemingly interchangeable actors. Set against this backdrop, my dissertation analyses the processes, norms, values and tensions that constitute data annotation for AI. In the first phase of this research, I have used an in-depth, qualitative approach to examine annotation practices in data annotation centres in India. This research has revealed the tensions and frictions involved in using human judgement to train and fine-tune AI models. Specifically, it has shown how the structures put in place in data work to mitigate human subjectivity (and the potential for harm), oversimplify the contexts being annotated and thus lack real-world efficacy. Building on this work, the next phase of my PhD investigates responsible approaches to dataset production that, rather than seeking to eliminate human subjectivity, actively incorporate it into AI development cycles. Through this work, I aim to develop a framework for responsible data annotation that will identify actionable opportunities to improve how machines learn from diverse human experiences.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**;
Collaborative and social computing.

KEYWORDS

data work, annotation, supply chains, global south, ai, ethnography

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1 BACKGROUND AND MOTIVATION

AI research and development has received a renewed boost over the last decade or so. Commercial enterprises, research labs, and governments alike are keen on leveraging and expanding AI's ability to

forecast, personalise, and automate. Within this hyped discourse on AI, much of the attention, resources and recognition, is garnered by the computational capabilities of AI. However, these capabilities are enabled by training AI models on large scale datasets that are rarely available readily. Data has to be collected or generated, cleaned, curated, classified, annotated and verified before AI-based systems can harness them for “intelligent” output. The significant effort involved in data cleaning, annotation, and verification is largely carried out by human data workers [3, 16]. However, there is far less recognition and consideration for the indispensable, and foundational role of human intervention in AI system building. This rarely considered human aspect of AI system building, often termed as data annotation or labelling, is the central focus of my research.

Data annotators not only provide clean, labelled data, they also validate the model output and stand-in for the AI system at points of breakdown or failure [27]. These tasks and their outcomes, too become labelled input datasets for subsequent rounds of training. Building training datasets is thus not a one-time activity that is conducted in the initial stages of AI development. Rather it is a continuous, ongoing process of sense-making [10] that accompanies and integrates within AI system building at different points.

Data annotation work is primarily carried out in regions in South Asia, Latin America and Africa [12], and typically framed as impact sourcing, a specific type of business process outsourcing (BPO) leveraging the AI-driven demand for data work as an entry point for workers from marginalised communities and regions with limited alternative employment opportunities to join the global tech supply chains [6]. It reflects a modernist discourse of progress, that tapping into the tech sector is a chance to establish links to the centres of power in the global north. Further, it advances a notion of seamless integration and collapsing distances within AI supply chains that draws from the globalisation rhetoric of free flow of ideas, goods, people and resources.

Despite data annotation being integral to AI systems, the prevailing model-centric view of AI celebrates and accords high status to algorithmic aspects of AI [19]. This overt emphasis on models overshadows the contributions made by people and, more worryingly, erases those drafted into providing data support for these systems [18]. Apart from their contributions being rarely acknowledged, and their work being framed as mundane and low-skilled (Irani 2015), data annotators are also viewed as the source of bias in AI systems, for introducing their ‘subjectivity’ in providing labelling input. However, recent work in CSCW and HCI [5, 9, 15, 17, 30] expose how the power asymmetries in data annotation prevent data workers from introducing their own subjectivities or indeed, meaningfully engaging with the annotation tasks. Building on this scholarship, my research aims to study the actors, practices, norms, hierarchies, structures, and values that constitute data annotation

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work and through that, examine the effectiveness and fragility of AI.

2 RESEARCH OBJECTIVES

The rapidly growing AI industry is sustained by millions of workers engaged in data work through digital labour platforms [11] and outsourcing companies [10, 30] to sustain AI's research, development and use. Set against this backdrop, my research, in conversation with a growing body of critical research on data work [7, 9, 19, 30], examines the role of the human data worker in supporting AI system building. I am interested in understanding how their input — situated human judgement over socio-culturally specific contexts — feeds into AI system-building, what makes this possible and what its limits and exclusions might be. To this end, my research is guided by the following research questions:

RQ1: What actors, practices, places, norms, policies and values make up the process of creating training datasets for AI?

RQ2: How are these diverse entities interconnected and configured to support AI system-building?

RQ3: What are the implications of this configuration for the people involved in labelling as well as the AI systems they support?

RQ4: How might we intervene in this configuration to pave the way for just and responsible futures of data labelling for AI?

In examining the human labours that support and sustain AI development, I pay particular attention to how it is situated within the global AI supply chains. To investigate the interconnections within the AI supply chain and notions of seamless integration, I draw inspiration from critical studies of globalist projects and knowledge production regimes [2, 8, 26, 28]. In particular, Anna Tsing's critique [25, 26] of universalist claims and attention to frictions that make up trans-national scales and global flows are relevant and insightful to analyse data labours in AI. In following the on-ground practices of data annotation and their interconnections with other aspects of AI development, I focus on the contingent norms, histories and structures that constitute annotation work and the supply chains of AI production.

3 OVERVIEW OF EMPIRICAL STUDIES

In operationalizing the research objectives outlined above, I adopt immersive, qualitative methods — especially ethnography — to examine the practices, people and structures that constitute data annotation. Here, I draw inspiration from a long tradition of inductive approaches and ethnographic research in CSCW [13, 14, 23, 24, 29] that focuses on how actors order their activities and draw on artefacts, technologies and their knowledge, skills and expertise to conduct 'work'. In the sections that follow, I outline the empirical studies conducted as part of this research project. In my empirical work, I focus on India as the primary site for this research, both for its prominence in the annotation industry, and a growing interest in the region to build datasets and AI for 'low resource' Indian languages.

3.1 Data annotation on the ground

In the first phase of the PhD, I conducted an in-depth, ethnographic study of annotation practices at data centres in 2022 where I observed annotation projects in domains such as retail shopping,

ecommerce, navigation, and security. I undertook this project with the goal of uncovering the structures, norms, and tensions that constitute globalised dataset production, and consequently impact the outcomes of AI systems.

Findings from the study show that dataset production for AI is characterised by the presumption of total countability - the notion that everything, from tasks, datasets and deliverables, to workers, work time, quality and performance, can be managed by applying the logics of counting [1]. These logics constitute annotation tasks as reductive, homogenous and standardised, diminishing the importance of human judgement or discretion needed and indeed sought after in many annotation tasks. This could have adverse effects particularly for issues such as bias, fairness, diversity, and representation that are themselves often rendered through the very same enumerating logics. This study reveals the restricted scope within which human discretion is sought and used to train AI systems under prevailing approaches to annotation.

3.2 Towards ethically responsible annotation practice

The insights from the above study highlight the deep fissures in how annotation and dataset production is integrated within the AI production pipeline. This informs the next phase of research exploring alternative approaches to dataset production for AI that foster and value the participation and contributions of data workers. I explore this in an ongoing study in which I ethnographically follow the dataset production activities of an open source, civic-tech organisation that develops ML tools and datasets to detect and redact gender abusive content online in Indian languages. Here, what constitutes abusive content is defined and annotated by working closely with people of minoritized genders with situated knowledge of online gendered abuse. In this study, I engage with the organisation's feminist and community-oriented approach to annotation practice, particularly the socio-technical choices, challenges and constraints in placing the diverse experiences and knowledge of the annotators as central to dataset and ML development.

Approaching annotation through a feminist lens also requires confronting how the expertise and contributions of annotators should be recognised and valued within AI development. I will examine this question in the context of the civic tech organisation's work through interviews and workshops with a variety of stakeholders including annotators, data requesters, AI for social good projects, responsible AI experts, and feminist scholars, to unpack the costs, tensions and trade-offs underlying dataset production and governance. Findings will identify key considerations for valuing annotation work and provide guidelines that can be adopted by other AI projects. Through this work and in collaboration with the civic tech project, I aim to define a best practices framework that forms the basis for responsible data annotation in AI.

4 EXPECTED CONTRIBUTIONS

Amidst a surging interest in AI adoption, there is an urgent need to deepen our understanding of the work involved in dataset production for AI development as the wider challenges faced at this layer may have serious implications for the AI systems we build. Recent work within CSCW and HCI has begun paving the way to

understand worker perspectives and experiences in data and AI production [4, 7, 9, 20, 30]. My research engages with this line of enquiry, contributing crucial insights into current structures and practices of annotation, and exploring ways to engage in more responsible dataset production.

Through the empirical studies of annotation work, my research contributes a rich, in-depth account of the day-to-day work practices of data annotators. In this, my work engages closely with the themes of power and authority developed in earlier CSCW work [9, 19, 30] and extends these themes by analysing the interconnections within AI production and the unequal, diverging interests and encounters upon which it is contingent. I aim to highlight the serious limits and fissures in the prevailing approaches to data production that will have a bearing on the futures of AI- and data-driven work, how such work (and workers) get valued, and the broader efforts to introduce accountable and more just AI systems.

Recognising the issues impacting annotation and dataset production, my ongoing study explores alternative annotator-centric approaches to data production. Through this work, I aim to define best practices and guidelines for meaningful engagement with and just reward of annotation work. This will provide the basis for innovation in data annotation and feed into a framework to guide responsible dataset development and governance. Further, this research offers insights into Responsible AI practices and challenges in/from the majority world. This has not yet received much focus or analytical attention in the scholarship on Responsible AI and AI ethics that have been predominantly focussed on Western, Educated, Industrialised, Rich, and Democratic (or 'WEIRD') societies [21, 22].

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